

Wave-Optics Gravitational Lensing of Gravitational Waves in a Hierarchical Triple

Results

Final project — Gravitational Waves and AI-Assisted Research

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Abstract

We compute how a gravitational wave (GW) from a compact binary is lensed, in wave optics, by a third, more massive body in the same hierarchical triple, for the same triple at two epochs of the inner binary’s inspiral: a late chirp (lens frozen for the whole observation) and an earlier, quasi-monochromatic phase (the outer orbit’s motion is resolved, producing periodic “repeated lensing”). All figures and numbers here are produced by the scripts in `cases/`, checked against independent methods in `tests/`, and derived from first principles in `theory/theory.pdf`, which this document does not repeat. This page/PDF is the results; the repository behind it (`README.md`) is what makes them reproducible.

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1 The system

One hierarchical triple (`src/gwlens/system.py`), followed at two points in its own evolution:

inner binary (the source)	$m_1 = 20 M_\odot, m_2 = 15 M_\odot$ ($\mathcal{M} = 15.05 M_\odot$)
lens	$M_L = 5 \times 10^4 M_\odot$, at $D_L = 5$ kpc
outer orbit	$a_{\text{out}} = 1.82$ AU, $P_{\text{out}} = 4$ d, $i_{\text{out}} = 87^\circ$, circular

The lens mass is chosen (not astrophysically typical) so that the wave-optics parameter $w = 8\pi G M_L f / c^3$ is not extreme at either epoch’s frequency; the hierarchy $a_{\text{in}} \ll a_{\text{out}}$ holds by 2–6 orders of magnitude at both. Full justification: `theory/theory.pdf` Ch. 1, §6.1; regime checks: `tests/test_system.py`.

2 Case A — the chirp, static lens

The inner binary inspirals from $f = 10$ Hz to $f_{\text{isco}} = 125.6$ Hz in **15.18 s** — 4.4×10^{-5} of the 4-day outer period, so the lens sits at one impact parameter, $y_A = 1.589$ (the phase of closest projected approach during the outer orbit), for the entire chirp. The unlensed waveform

is the restricted TaylorF2 (2PN) frequency-domain post-Newtonian inspiral — the functional form used in real LIGO/Virgo matched-filter search templates, not a hand-built approximation (`src/gwlens/taylorf2.py`).

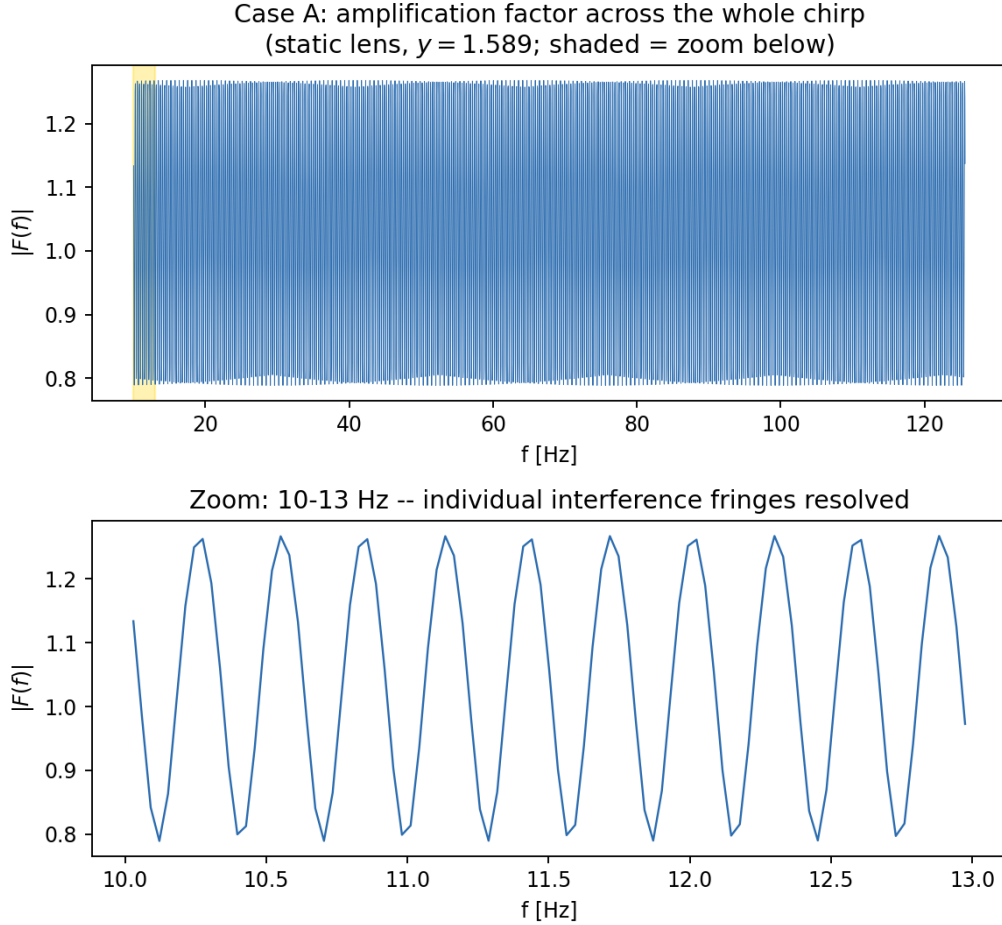


Figure 1: The amplification factor $F(f)$ across the whole chirp band (top) and zoomed to 3 Hz (bottom), where the individual two-image interference fringes are resolved. w spans 62 (band start) to 778 (merger) here — well above where `tests/test_waveoptics.py` validates the fast geometric-optics evaluator against the exact closed form to $< 4\%$, so this is genuinely wave interference, just too densely packed to resolve at the full-band scale: the checked transition toward geometric optics as merger approaches.

Peak-sample amplification is 1.038; RMS amplification (averaged over the whole chirp) is 1.056. These being close but not equal, despite $|F|$ ranging 0.79–1.27 across the band, is itself informative: $F(f)$ modulates *phase* as well as amplitude, so the lensed waveform is a *dephased* copy of the unlensed one, not simply a rescaled one — a genuinely wave-optics statement (in the geometric-optics limit, lensing *is* just a real magnification).

3 Case B — quasi-monochromatic, moving lens

Earlier in the same inspiral: $f = 0.05$ Hz, drifting only 4% over a 24-day (6 outer period) observation, with 60.2 outer periods still left before merger — quasi-monochromatic. $w_B = 0.309$ throughout (fixed; only the impact parameter $y(t)$ varies): squarely in the diffraction-dominated regime.

Because the lens and source are in the *same* system rather than at the usual well-separated cosmological distances, $D_{LS}(t)$ is set by the outer orbit itself and vanishes twice per period;

Case A: extended amplification pattern on the source plane
 int lens: axisymmetric rings; central bright spot = Einstein-ring diffraction spike; source at fixed $y_A = 1.589$, mark

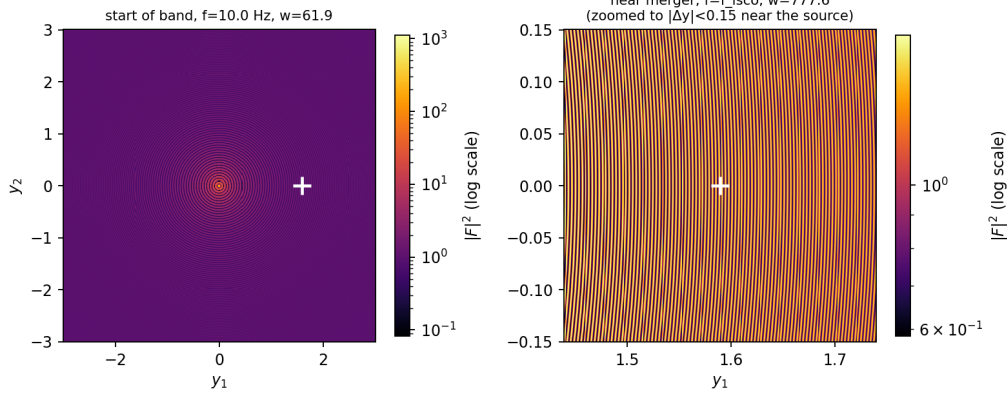


Figure 2: The extended amplification pattern on the source plane: concentric diffraction rings around the lens with a central Einstein-ring bright spike (finite in wave optics, where the geometric-optics magnification formally diverges — the GW analogue of the optical Arago/Poisson spot). Left: full pattern at band start ($w = 61.9$). Right: zoomed to $|\Delta y| < 0.15$ near merger ($w = 777.6$) — the same rings, now so closely spaced they look like near-straight fringes at this zoom.

exactly half the orbit turns out to be unlensed (source in front of the lens, not behind it, so no lensing geometry exists there at all — not an approximation, $F = 1$ by construction). This is the origin of the periodic pulses above, the wave-optics analogue of the “repeated lensing” found in the geometric-optics limit by D’Orazio and Loeb (2020). Full derivation: [theory/theory.pdf](#) §4.2–4.3 (which also quantifies exactly how far this pushes the standard thin-lens approximation, and shows the effect is confined to a ~ 60 s window per crossing, coincident with the already-trivial $F \approx 1$ region — not a concern for anything shown here).

4 What the two cases show together

The same lens, the same formalism, two very different-looking signatures, governed entirely by one timescale ratio (merger duration vs. outer period vs. observation length):

- Case A: a single, frozen diffraction pattern sampled at one point, swept across in *frequency* by the chirp itself — dense interference fringes, checked to approach the geometric-optics limit as $w \rightarrow \infty$.
- Case B: the same kind of pattern, at fixed frequency, sampled in *time* by the outer orbital motion — periodic pulses, with exactly half the orbit unlensed by the triple’s own geometry.
- Both cases: lensing is not simply “make it brighter.” $F(f)$ (or $F(t)$) is complex, and its phase leaves a real, checkable mark on the waveform beyond an overall amplitude.

5 Scope and limitations

Stated explicitly, not left implicit (each also logged in [wiki/log.md](#) at the point it was decided):

- Point-mass lens only; no extended mass distribution.
- TaylorF2 truncated at 2PN (Case A); Case B’s underlying frequency evolution is leading-order only.

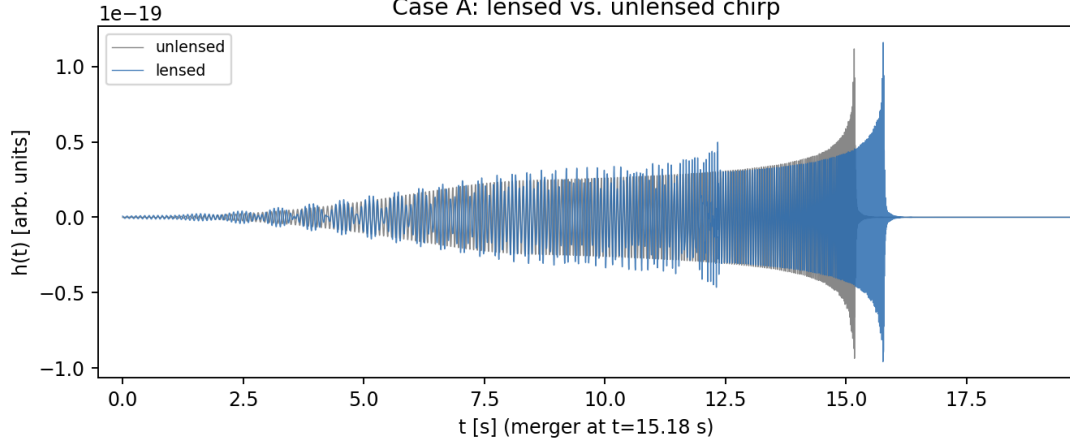


Figure 3: Lensed vs. unlensed time-domain chirp strain (arbitrary units, $D_{\text{eff}} = 5$ kpc — an illustratively loud, nearby hypothetical event, not a rate-realistic distance).

- No absolute strain calibration (inclination/polarization prefactors folded into an $O(1)$ constant) — only relative amplitudes and the lensing amplification factor are physical claims here.
- The idealized detector view has no noise, antenna pattern, or matched filtering — illustrative only, as scoped from the start.
- The thin-lens/paraxial approximation is quantifiably pushed near each Case B front/back crossing (§3); shown not to affect anything reported.

References

Daniel J. D’Orazio and Abraham Loeb. Repeated gravitational lensing of gravitational waves in hierarchical black hole triples. *Physical Review D*, 101:083031, 2020. doi: 10.1103/PhysRevD.101.083031.

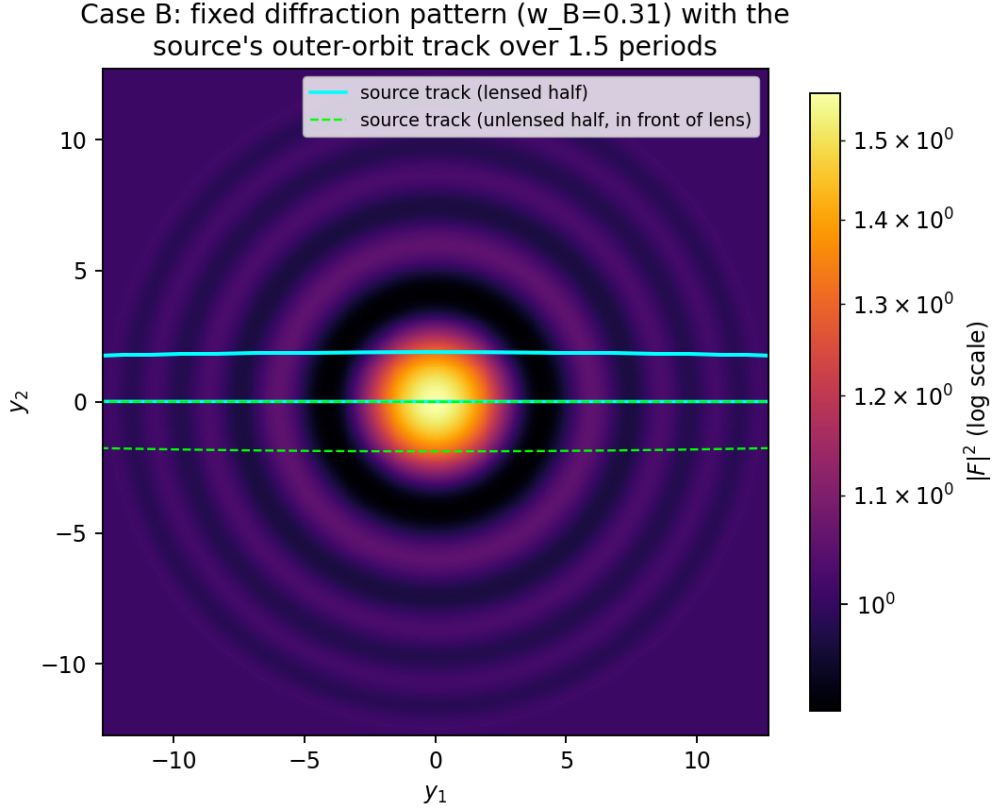


Figure 4: The same kind of fixed diffraction pattern as Case A (here at the much smaller $w_B = 0.309$) with the source's actual sky-projected outer-orbit track overlaid: cyan where the source is lensed (behind the lens along the line of sight), dashed green where it is not (in front of it — see below).

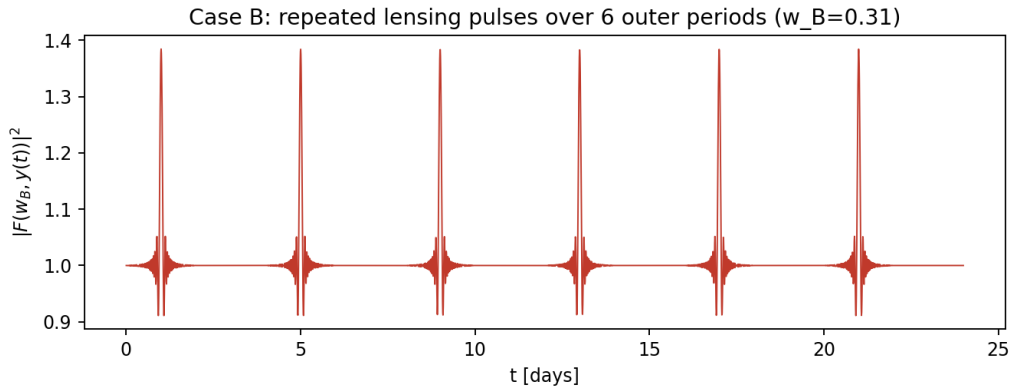


Figure 5: $|F(w_B, y(t))|^2$ over the full 24-day observation: six periodic amplification pulses, one per outer period, peaking at $|F|^2 = 1.385$, with diffraction ringing visible on both sides of each peak.

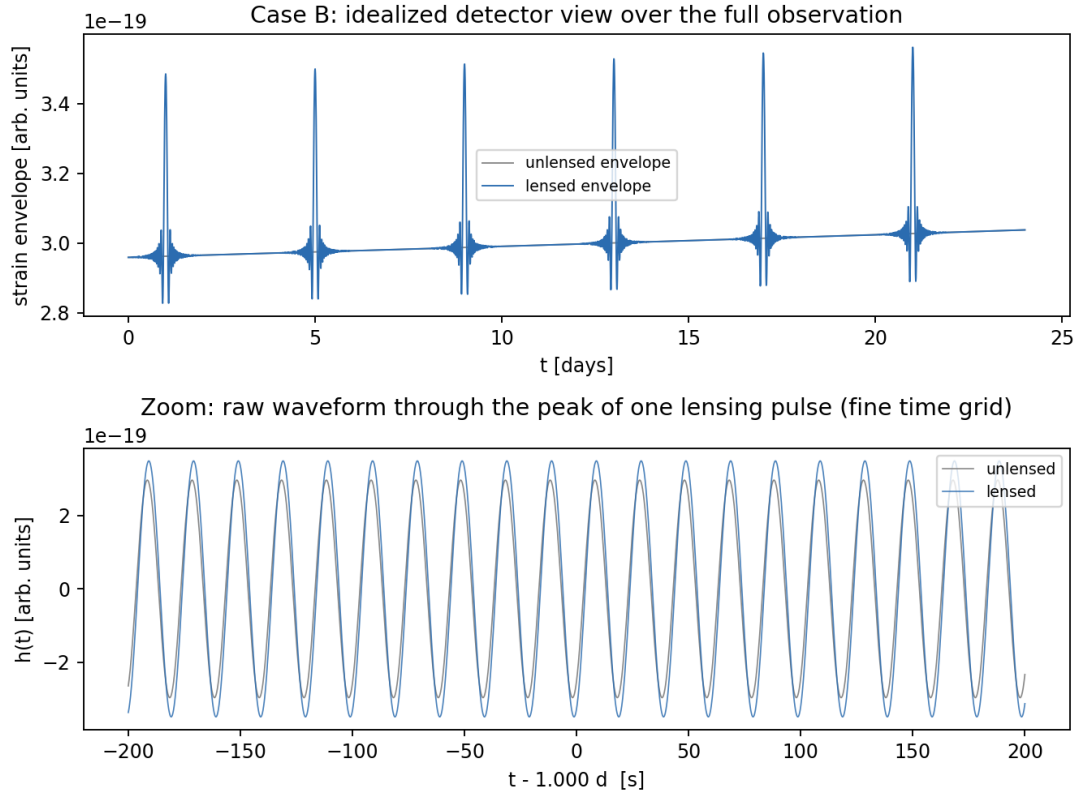


Figure 6: Idealized detector view. Top: strain envelope over the full observation, six clean pulses riding on the slowly-drifting carrier. Bottom: a fine time grid zoomed on one pulse peak, where the lensed and unlensed waveforms visibly dephase over a handful of carrier cycles.